

EDC notes

Difference between JFET and BJT.

Difference between BJT and JFET	
BJT (Bipolar Junction Transistor)	JFET (Junction Field Effect Transistor)
Current-controlled device	Voltage-controlled device
Uses both electrons and holes (bipolar)	Uses only majority carriers (unipolar)
Requires base current to operate	Gate current is almost zero
Low input impedance	Very high input impedance
Higher power consumption	Low power consumption
Better current gain	Better voltage control
More noise compared to JFET	Low noise operation
Control is less stable (temperature sensitive)	More temperature stable
Faster switching in some cases	Slower than BJT in many applications
Used in amplifiers, switching circuits	Used in buffer stages, low-noise amplifiers

JFET Parameters (Definition & Explanation)		
Parameter	Definition	Explanation / Formula
Drain Current (I_D)	Current flowing from drain to source	Controlled by gate voltage; main output current of JFET
Gate-Source Voltage (V_{GS})	Voltage between gate and source	Controls the channel width and hence I_D
Drain-Source Voltage (V_{DS})	Voltage between drain and source	Drives current through the channel
Zero-Gate Voltage Drain Current (I_{DSS})	Maximum drain current when $V_{GS} = 0$	$I_D = I_{DSS}$ (maximum conduction state)
Pinch-off Voltage (V_P)	Gate-source voltage where channel closes	At $V_{GS} = V_P$, $I_D \approx 0$
Transconductance (g_m)	Change in drain current with gate voltage	$g_m = \frac{\Delta I_D}{\Delta V_{GS}}$
Amplification Factor (μ)	Ratio of change in drain voltage to gate voltage effect	$\mu = g_m \times r_d$
Drain Resistance (r_d)	Small-signal output resistance	$r_d = \frac{\Delta V_{DS}}{\Delta I_D}$
Cut-off Voltage	Gate voltage where current becomes zero	Same as pinch-off in many cases



1. JFET Biasing Techniques

A. Fixed Biasing

This is the simplest form of biasing, where a dedicated external DC voltage source (V_{GG}) is applied directly to the gate to keep it at a negative potential relative to the source.

- **Advantages:** * Extremely simple circuit with minimal components.
 - Independent control over the gate voltage.
- **Disadvantages:**
 - **Poor stability:** The Q-point is highly dependent on individual JFET parameters (I_{DSS} and V_p), which vary drastically even between variations of the same manufacturing batch.
 - Requires two separate DC power supplies (V_{DD} and V_{GG}), which is inefficient for hardware design.

B. Self-Biasing

Self-biasing drops the need for a second power supply by utilizing a resistor (R_S) at the source terminal. The drain current I_D flowing through R_S creates a voltage drop ($V_S = I_D R_S$). Since the gate is grounded through a high-value resistor (R_G), the gate voltage is 0 V, making $V_{GS} = -I_D R_S$.

- **Advantages:**
 - Requires only a single power supply (V_{DD}).

- Provides better Q-point stability than fixed bias; if I_D tries to increase, V_S rises, making V_{GS} more negative, which naturally throttles I_D back down.
- **Disadvantages:**
 - Stability is still moderately affected by JFET parameter variations.
 - The source resistor introduces negative AC feedback, which reduces voltage gain unless a bypass capacitor (C_S) is added.

C. Voltage-Divider Biasing

This technique uses a resistive divider (R_1 and R_2) connected across V_{DD} to establish a fixed positive voltage at the gate (V_G). A source resistor (R_S) is still used to keep the source voltage (V_S) higher than V_G , ensuring that $V_{GS} = V_G - V_S$ remains negative.

- **Advantages:**
 - **Excellent Q-point stability:** It is the most stable biasing method against temperature fluctuations and manufacturing variations in I_{DSS} and V_p .
 - Operates on a single power supply.
- **Disadvantages:**
 - Higher component count (requires more resistors).
 - The input impedance is slightly reduced because the AC signal sees $R_1 \parallel R_2$ instead of just one massive R_G .

Reasoning question

Best JFET Biasing Method (Q-point Stability)

From the given options:

👉 Voltage Divider Bias provides the best Q-point stability.

Why Voltage Divider Bias is Best?

In voltage divider bias:

- Gate voltage is set by a resistor divider network
- Source resistor provides **negative feedback**
- Both V_{GS} and I_D become much more stable

Stability mechanism:

If I_D increases:

- V_S increases (due to R_S)
- V_{GS} decreases (more negative)
- This reduces I_D back to normal

👉 So both **gate control** + **source feedback** stabilize the Q-point.

How to improve stability of self bias circuit? How can the drain current in a self-bias circuit be made more constant?**Reasoning question**

Self bias JFET circuit is mainly achieved by:

✔ Using a Source Resistor (Self-bias with Source Feedback)

Correct Method (as per Floyd)

👉 Increase source resistance R_S to provide degenerative (negative) feedback

Why this works (Floyd explanation style)

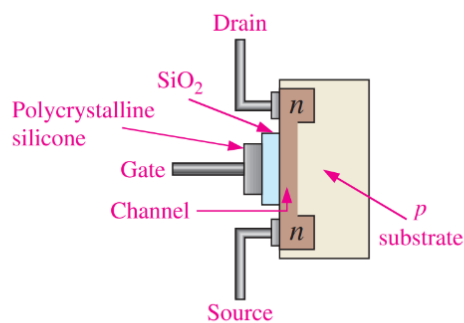
If drain current I_D increases:

- Voltage across R_S increases $\rightarrow V_S$ increases
- So $V_{GS} = V_G - V_S$ becomes more negative
- This reduces I_D back toward its original value

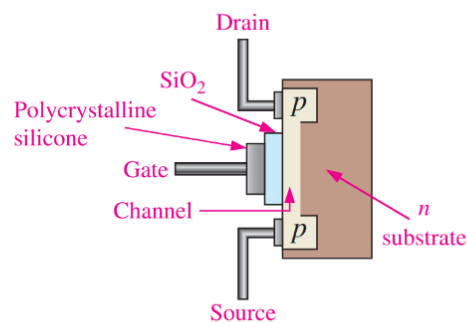
👉 This feedback action stabilizes the Q-point.

8-5 THE MOSFET

The **MOSFET** (metal oxide semiconductor field-effect transistor) is another category of field-effect transistor. The MOSFET, different from the JFET, has no pn junction structure; instead, the gate of the MOSFET is insulated from the channel by a silicon dioxide (SiO_2) layer. The two basic types of MOSFETs are enhancement (E) and depletion (D). Of the two types, the enhancement MOSFET is more widely used. Because polycrystalline silicon is now used for the gate material instead of metal, these devices are sometimes called IGFETs (insulated-gate FETs).



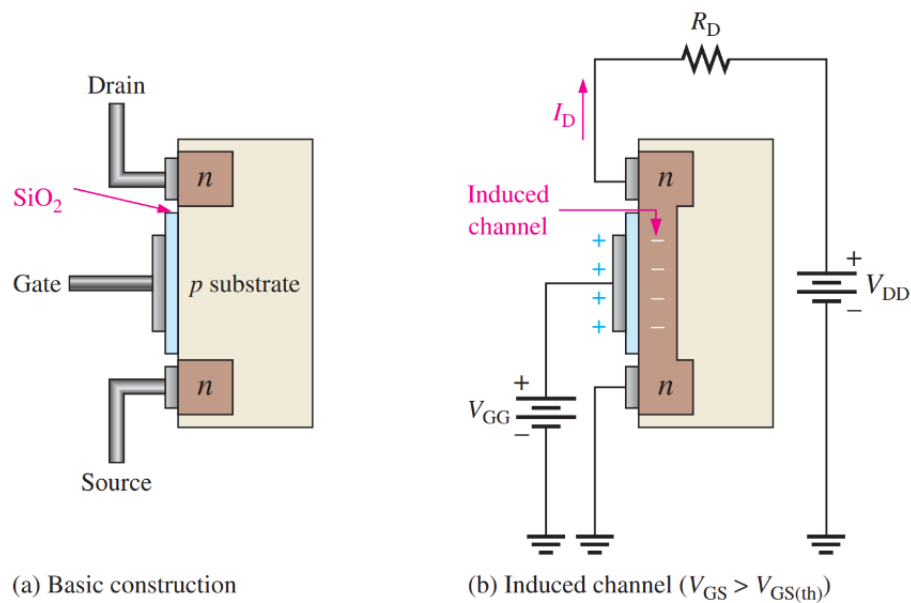
(a) n channel



(b) p channel

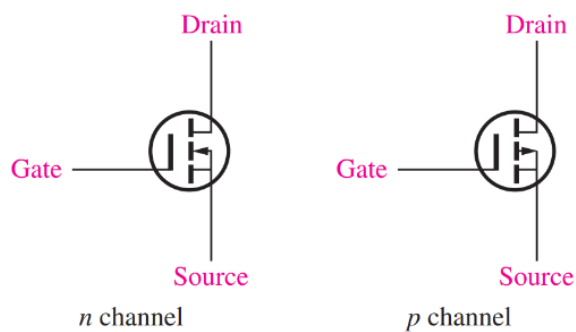
Enhancement MOSFET (E-MOSFET)

The E-MOSFET operates *only* in the enhancement mode and has no depletion mode. It differs in construction from the D-MOSFET, which is discussed next, in that it has no structural channel. Notice in Figure 8–34(a) that the substrate extends completely to the SiO_2 layer. For an n -channel device, a positive gate voltage above a threshold value *induces* a channel by creating a thin layer of negative charges in the substrate region adjacent to the SiO_2 layer, as shown in Figure 8–34(b). The conductivity of the channel is enhanced by increasing the gate-to-source voltage and thus pulling more electrons into the channel area. For any gate voltage below the threshold value, there is no channel.



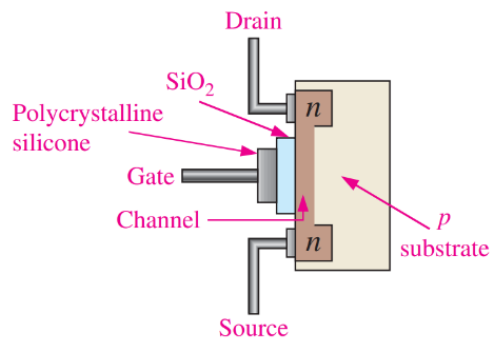
► **FIGURE 8–35**

E-MOSFET schematic symbols.

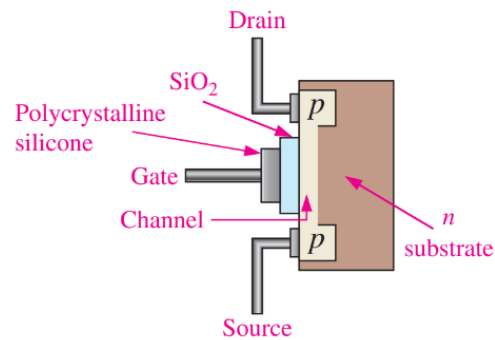


Depletion MOSFET (D-MOSFET)

Another type of MOSFET is the depletion MOSFET (D-MOSFET), and Figure 8–36 illustrates its basic structure. The drain and source are diffused into the substrate material and then connected by a narrow channel adjacent to the insulated gate. Both n -channel and p -channel devices are shown in the figure. We will use the n -channel device to describe the basic operation. The p -channel operation is the same, except the voltage polarities are opposite those of the n -channel.



(a) n channel



(b) p channel

1. N-Channel Enhancement-Mode (Normally-Off)

- **Substrate (Body):** P-type silicon.
- **Source & Drain:** Two separate, heavily doped N-type wells.
- **The Channel: None.** There is a physical gap of P-type silicon separating the two N-type wells.
- **How it works:** You must apply a positive voltage to the gate to attract electrons, creating a temporary N-channel across the gap.

2. P-Channel Enhancement-Mode (Normally-Off)

- **Substrate (Body):** N-type silicon.
- **Source & Drain:** Two separate, heavily doped P-type wells.
- **The Channel: None.** The N-type substrate completely separates the two P-type wells.
- **How it works:** You must apply a negative voltage to the gate to attract holes, creating a temporary P-channel across the gap.

3. N-Channel Depletion-Mode (Normally-On)

- **Substrate (Body):** P-type silicon.
- **Source & Drain:** Two heavily doped N-type wells.
- **The Channel: Built-in.** A thin, physical strip of N-type silicon permanently connects the source well to the drain well during manufacturing.
- **How it works:** Current flows freely through this built-in bridge until you apply a negative gate voltage to push the electrons away and choke off the channel.

4. P-Channel Depletion-Mode (Normally-On)

- **Substrate (Body):** N-type silicon.
- **Source & Drain:** Two heavily doped P-type wells.
- **The Channel: Built-in.** A thin, physical strip of P-type silicon permanently bridges the source and drain wells.
- **How it works:** Current flows freely until you apply a positive gate voltage to push the holes away and empty the channel.

now explain two modes of depletion and basing

The two modes of a Depletion-mode MOSFET refer to the two different ways you can operate the device: **Depletion Mode** (turning it off) and **Enhancement Mode** (turning it up).

Unlike Enhancement-only MOSFETs, a Depletion MOSFET is highly versatile because it can operate in both modes depending on the gate bias (voltage) you apply.

The Two Operating Modes

1. Depletion Mode (Depleting the Channel)

- **The Goal:** Reduce or stop the current flow (turn the switch "OFF").
- **The Action:** You apply a voltage that **repels** the natural charge carriers out of the built-in channel.
- **The Result:** The channel empties out (depletes), narrows down, and eventually chokes off entirely. No current can pass.

2. Enhancement Mode (Enhancing the Channel)

- **The Goal:** Increase the current flow beyond its natural state (turn the switch "UP").
- **The Action:** You apply a voltage that **attracts** even more charge carriers into the built-in channel.
- **The Result:** The channel becomes wider and more conductive than it was at rest, allowing much more current to flow.

How "Basing" (Biasing) Works For Both Types

Biasing means applying the correct polarity of voltage to the gate relative to the source (V_{GS}) to achieve these modes.

N-Channel Depletion MOSFET (Built-in N-channel, P-type substrate)

- **At Zero Bias ($V_{GS} = 0V$):** The built-in N-channel is open. Current flows naturally.
- **Depletion Mode Biasing ($V_{GS} = \text{Negative}$):** Pushing negative voltage into the gate repels the negative electrons out of the N-channel. This creates a depletion region, narrowing the path until current stops entirely (at the "pinch-off" voltage).
- **Enhancement Mode Biasing ($V_{GS} = \text{Positive}$):** Pushing positive voltage into the gate pulls extra electrons from the P-substrate into the channel. This widens the path and boosts current flow.

P-Channel Depletion MOSFET (Built-in P-channel, N-type substrate)

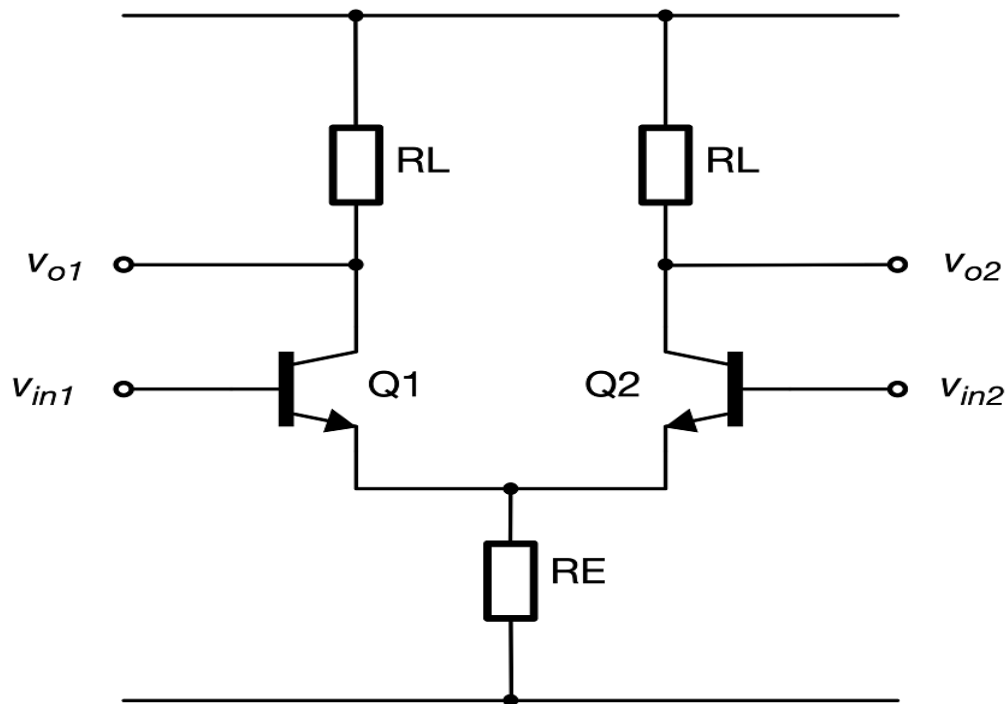
- **At Zero Bias ($V_{GS} = 0V$):** The built-in P-channel is open. Current flows naturally.
- **Depletion Mode Biasing ($V_{GS} = \text{Positive}$):** Pushing positive voltage into the gate repels the positive holes out of the P-channel, choking off the current.
- **Enhancement Mode Biasing ($V_{GS} = \text{Negative}$):** Pushing negative voltage into the gate pulls extra holes from the N-substrate into the channel, widening the path and boosting current.

Biasing Quick Reference

Channel Type	To Deplete (Turn Off)	To Enhance (Boost)
N-Channel Depletion	Negative Gate Voltage ($- V_{GS}$)	Positive Gate Voltage ($+ V_{GS}$)
P-Channel Depletion	Positive Gate Voltage ($+ V_{GS}$)	Negative Gate Voltage ($- V_{GS}$)

CHAPTER 12

Differential amplifier in mosfet?



A **MOSFET differential amplifier** is a circuit that amplifies the **difference between two input signals** while rejecting signals common to both inputs (noise/common-mode signals).

It is widely used in:

- Operational amplifiers (Op-amps)
- Comparators
- Audio circuits
- Analog ICs

Basic MOSFET Differential Amplifier

Usually two identical MOSFETs are used.

Main Parts

- **M1 and M2** → identical MOSFETs
- **RD1 and RD2** → drain resistors
- **VDD** → supply voltage
- **Vin1 and Vin2** → input signals
- **Current source or resistor** at source terminals

Working Principle

Case 1: Both Inputs Equal

If:

[

$$V_{in1} = V_{in2}$$

]

then:

- Both MOSFETs conduct equally
- Drain voltages remain equal
- Output difference is nearly zero

This removes common noise.

Case 2: $V_{in1} > V_{in2}$

If:

[

$$V_{in1} > V_{in2}$$

]

then:

- M1 conducts more current
- Voltage drop across R_{D1} increases
- Drain voltage of M1 decreases

At the same time:

- M2 conducts less
- Drain voltage of M2 increases

So the amplifier produces a large output difference.

Differential Output

Output is usually:

[

$$V_{\text{out}} = V_{\text{o1}} - V_{\text{o2}}$$

]

where:

- (V_{o1}) = drain voltage of M1
- (V_{o2}) = drain voltage of M2

Important Characteristics

Parameter	Meaning
Differential Gain	Amplification of difference between inputs
Common Mode Gain	Gain for common signals
CMRR	Ability to reject common noise
Input Resistance	Very high in MOSFETs
Power Consumption	Low

Advantages of MOSFET Differential Amplifier

- Very high input impedance
- Low power consumption
- Good noise rejection
- Easy IC fabrication
- High voltage gain

Applications

Application	Use
Op-Amps	Input stage
Comparators	Compare voltages
Audio Systems	Noise reduction
ADCs	Signal conditioning
Communication Circuits	Differential signaling

Simple Explanation

Think of it like a “difference detector”:

- If both inputs are same → almost no output

- If inputs differ → amplifier responds strongly

In which circuit differential amplifiers are used?

The **most common circuit** where a differential amplifier is used is the **operational amplifier (Op-Amp)**.

How it is used in Op-Amp:

Inside every op-amp, the first stage is a **differential amplifier made using MOSFETs or BJTs**. It takes two input voltages:

- Non-inverting input (+)
- Inverting input (-)

Then it amplifies only the **difference between them**.

Why it is important:

- It gives the op-amp its ability to compare signals
- It removes unwanted noise that appears on both inputs (common-mode noise)
- It ensures accurate amplification of small signals

Simple idea:

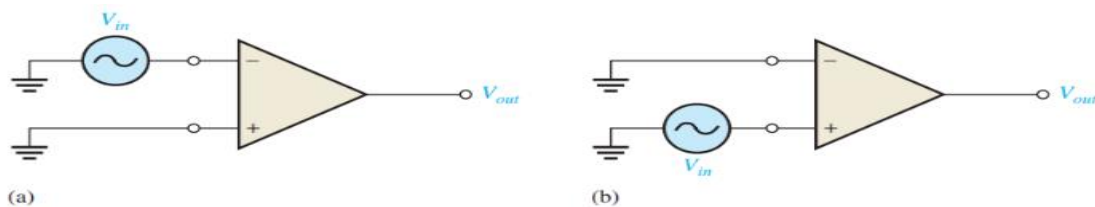
If both inputs have the same noise, the differential amplifier cancels it, and only the real difference is amplified. That's why it is the **core and most commonly used circuit in op-amps**.

Differential mode and common mode?

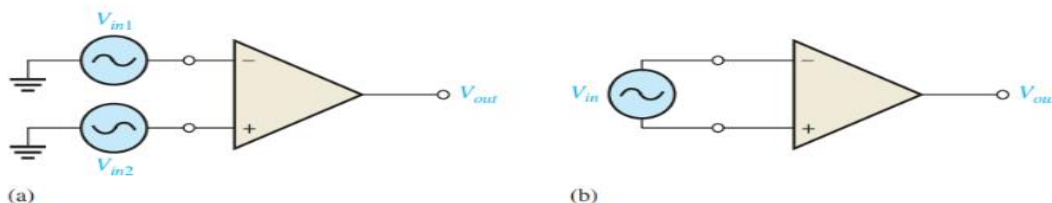
Input Signal Modes

Recall that the input signal modes are determined by the differential amplifier input stage of the op-amp.

Differential Mode In the **differential mode**, either one signal is applied to an input with the other input grounded or two opposite-polarity signals are applied to the inputs. When an op-amp is operated in the single-ended differential mode, one input is grounded and a signal voltage is applied to the other input, as shown in Figure 12–4. In the case where the signal voltage is applied to the inverting input as in part (a), an inverted, amplified signal voltage appears at the output. In the case where the signal is applied to the non-inverting input with the inverting input grounded, as in Figure 12–4(b), a noninverted, amplified signal voltage appears at the output.

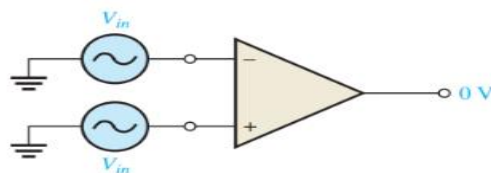


In the double-ended differential mode, two opposite-polarity (out-of-phase) signals are applied to the inputs, as shown in Figure 12–5(a). The amplified difference between the two inputs appears on the output. Equivalently, the double-ended differential mode can be represented by a single source connected between the two inputs, as shown in Figure 12–5(b).



Common Mode In the **common mode**, two signal voltages of the same phase, frequency, and amplitude are applied to the two inputs, as shown in Figure 12–6. When equal input signals are applied to both inputs, they tend to cancel, resulting in a zero output voltage.

► **FIGURE 12–6**
Common-mode operation.



This action is called *common-mode rejection*. Its importance lies in the situation where an unwanted signal appears commonly on both op-amp inputs. Common-mode rejection means that this unwanted signal will not appear on the output and distort the desired signal. Common-mode signals (noise) generally are the result of the pick-up of radiated energy on the input lines, from adjacent lines, the 60 Hz power line, or other sources.

Operational amplifier practical model and ideal model?

Ideal vs. Practical Op-Amp Comparison	
Ideal Op-Amp	Practical Op-Amp
Infinite Open-Loop Gain ($A_v = \infty$): Implies that even a microscopic voltage difference between the inputs results in a large output voltage.	Finite Open-Loop Gain ($A_v \approx 10^5$ to 10^6): The internal gain is extremely high but limited by semiconductor physics.
Infinite Input Impedance ($Z_{in} = \infty$): It draws absolutely zero current from the signal source ($I_+ = I_- = 0$).	High Input Impedance ($Z_{in} \approx 2\text{ M}\Omega$ to $10^{12}\Omega$): It draws a microscopic leakage current, known as input bias current.
Zero Output Impedance ($Z_{out} = 0\ \Omega$): Can drive any load resistance without dropping output voltage.	Low Output Impedance ($Z_{out} \approx 10\ \Omega$ to $100\ \Omega$): Internal resistance causes minor voltage drops under heavy loads.
Infinite Bandwidth ($BW = \infty$): Amplifies any frequency signal flawlessly from DC (0 Hz) to infinitely high radio frequencies.	Finite Bandwidth: Gain drops off at higher frequencies. It is limited by an internal compensation capacitor (Gain-Bandwidth Product).
Infinite CMRR ($\text{CMRR} = \infty$): Perfectly rejects common noise signals present on both input terminals simultaneously.	High CMRR ($\text{CMRR} \approx 80\text{ dB}$ to 110 dB): Rejects most common-mode noise, but a tiny fraction of the noise still leaks to the output.
Infinite Slew Rate ($SR = \infty$): The output voltage can change instantaneously (0 ns response time) to track any sharp input edge.	Finite Slew Rate ($SR \approx 0.5\text{ V}/\mu\text{s}$ to $15\text{ V}/\mu\text{s}$): Output voltage slope is physically limited, causing distortion in fast square waves.

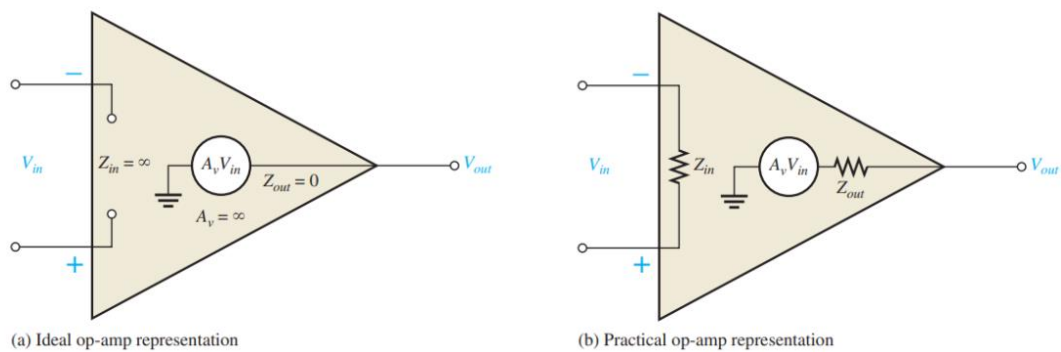
The Ideal Op-Amp

To illustrate what an op-amp is, let's consider its ideal characteristics. A practical op-amp, of course, falls short of these ideal standards, but it is much easier to understand and analyze the device from an ideal point of view.

First, the ideal op-amp has *infinite voltage gain* and *infinite bandwidth*. Also, it has an *infinite input impedance* (open) so that it does not load the driving source. Finally, it has a *zero output impedance*. Op-amp characteristics are illustrated in Figure 12–2(a). The input voltage, V_{in} , appears between the two input terminals, and the output voltage is $A_v V_{in}$, as indicated by the internal voltage source symbol. The concept of infinite input impedance is

introduction of the integrated circuit did the op-amp start to fulfill its true potential. In 1964, the first integrated circuit op-amp, designated the 702, was developed by Fairchild Semiconductor. This was later followed by the 709 and eventually the 741, which has become an industry standard.

a particularly valuable analysis tool for the various op-amp configurations, which will be discussed in Section 12–4.



Closed loop negative feedback:

Closed-Loop Voltage Gain, A_d

The **closed-loop voltage gain** is the voltage gain of an op-amp with external feedback. The amplifier configuration consists of the op-amp and an external negative feedback circuit that connects the output to the inverting input. The closed-loop voltage gain is determined by the external component values and can be precisely controlled by them.

Noninverting Amplifier

An op-amp connected in a **closed-loop** configuration as a **noninverting amplifier** with a controlled amount of voltage gain is shown in Figure 12–16. The input signal is applied to the noninverting (+) input. The output is applied back to the inverting (–) input through the feedback circuit (closed loop) formed by the input resistor R_i and the feedback resistor R_f . This creates negative feedback as follows. Resistors R_i and R_f form a voltage-divider circuit, which reduces V_{out} and connects the reduced voltage V_f to the inverting input. The feedback voltage is expressed as

$$V_f = \left(\frac{R_i}{R_i + R_f} \right) V_{out}$$

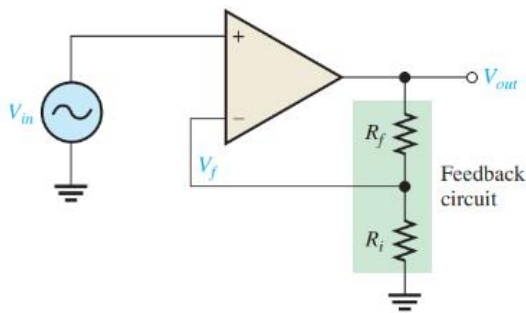


FIGURE 12–16
Noninverting amplifier.

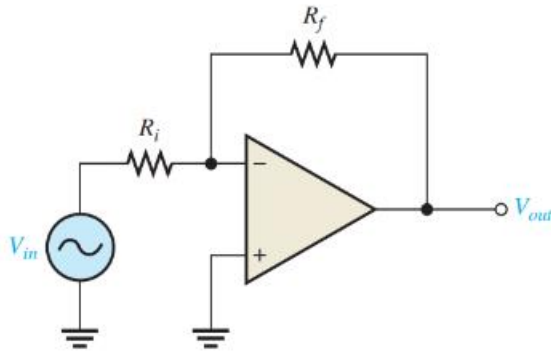
The difference of the input voltage, V_{in} , and the feedback voltage, V_f , is the differential input to the op-amp, as shown in Figure 12–17. This differential voltage is amplified by the open-loop voltage gain of the op-amp (A_{ol}) and produces an output voltage expressed as

$$V_{out} = A_{ol}(V_{in} - V_f)$$

Difference	
Inverting Amplifier	Non-Inverting Amplifier
Input at (–) terminal	Input at (+) terminal
Output is inverted (180° phase shift)	Output is in phase
Gain is negative	Gain is positive
Lower input impedance	Higher input impedance

Inverting Amplifier

An op-amp connected as an **inverting amplifier** with a controlled amount of voltage gain is shown in Figure 12–20. The input signal is applied through a series input resistor R_i to the inverting ($-$) input. Also, the output is fed back through R_f to the same input. The non-inverting ($+$) input is grounded.



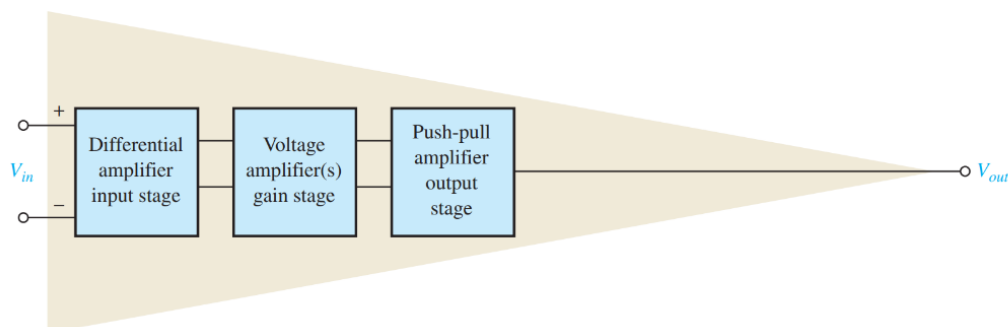
◀ **FIGURE 12–20**
Inverting amplifier.

The Practical Op-Amp

Although **integrated circuit (IC)** op-amps approach parameter values that can be treated as ideal in many cases, the ideal device can never be made. Any device has limitations, and the IC op-amp is no exception. Op-amps have both voltage and current limitations. Peak-to-peak output voltage, for example, is usually limited to slightly less than the two supply voltages. Output current is also limited by internal restrictions such as power dissipation and component ratings.

Characteristics of a practical op-amp are *very high voltage gain*, *very high input impedance*, and *very low output impedance*. These are labelled in Figure 12–2(b). Another practical consideration is that there is always noise generated within the op-amp. **Noise** is an undesired signal that affects the quality of a desired signal. Today, circuit designers are using smaller voltages that require high accuracy, so low-noise components are in greater demand. All circuits generate noise; op-amps are no exception, but the amount can be minimized.

Internal Block Diagram of an Op-Amp A typical op-amp is made up of three types of amplifier circuits: a differential amplifier, a voltage amplifier, and a push-pull amplifier, as shown in Figure 12–3. The **differential amplifier** is the input stage for the op-amp. It provides amplification of the difference voltage between the two inputs. The second stage is usually a class A amplifier that provides additional gain. Some op-amps may have more than one voltage amplifier stage. A push-pull class B amplifier is typically used for the output stage.



Explain the effect of negative feedback on the open-loop and closed-loop gain of an operational amplifier. Why is negative feedback used in op-amp circuits?

Effect of Negative Feedback on Open-loop and Closed-loop Gain

Copy message

1. Open-loop gain (without feedback)

- An operational amplifier has a **very large open-loop gain** (typically 10^5 to 10^6).
- Even a very small input difference produces a **very large output**, so it is unstable for direct use.

$$A_{OL} = \frac{V_{out}}{V_{in}}$$

2. Closed-loop gain (with negative feedback)

When **negative feedback** is applied, a portion of the output is fed back to the inverting input. This reduces the effective input difference.

- The overall gain becomes **controlled and stable**
- It depends mainly on external resistors, not internal op-amp properties

$$A_{CL} = \frac{A_{OL}}{1 + A_{OL}\beta}$$

Why Negative Feedback is Used in Op-Amp Circuits

Negative feedback is used because it improves overall performance:

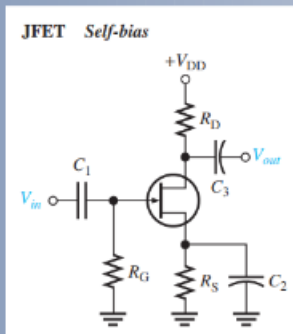
1. **Stabilizes gain**
 - Makes gain depend on resistors, not op-amp variations.
2. **Reduces distortion**
 - Output becomes more linear and accurate.
3. **Increases bandwidth**
 - Allows the amplifier to work over a wider frequency range.
4. **Improves stability**
 - Prevents saturation and uncontrolled output.
5. **Reduces noise sensitivity**
 - Minimizes effect of internal noise and disturbances.

Final Idea (Simple)

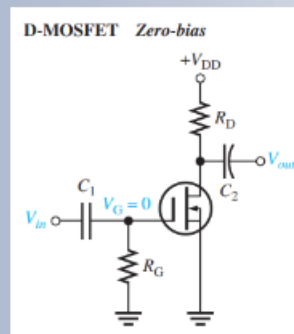
- Open-loop gain = very high but unusable
- Closed-loop gain = reduced but stable and useful

Chapter 9

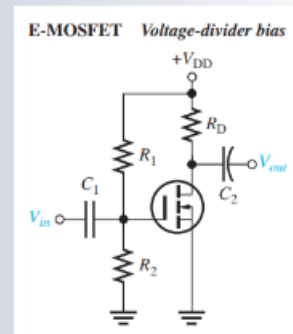
COMMON-SOURCE AMPLIFIERS



- $I_D = I_{DSS} \left(1 - \frac{I_D R_S}{V_{GS(off)}} \right)^2$
- $A_v = g_m R_d$
- $R_{in} = R_G \parallel \left(\frac{V_{GS}}{I_{GSS}} \right)$



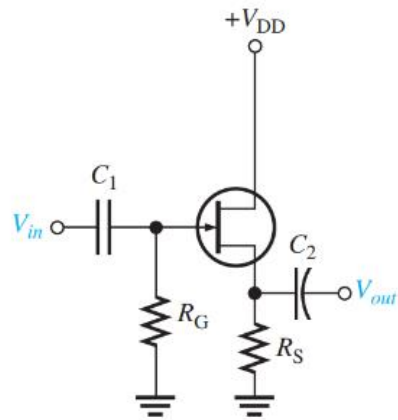
- $I_D = I_{DSS}$
- $A_v = g_m R_d$
- $R_{in} = R_G \parallel \left(\frac{V_{GS}}{I_{GSS}} \right)$



- $I_D = K(V_{GS} - V_{GS(th)})^2$
- $A_v = g_m R_d$
- $R_{in} = R_1 \parallel R_2 \parallel \left(\frac{V_{GS}}{I_{GSS}} \right)$

COMMON-DRAIN AMPLIFIER

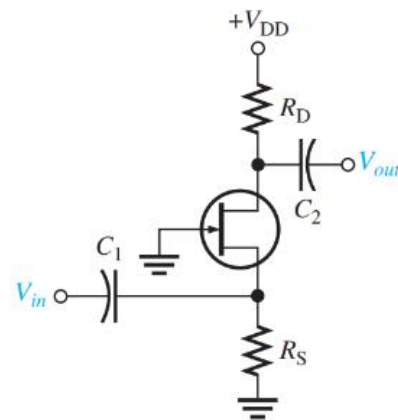
JFET Self-bias



- $I_D = I_{DSS} \left(1 - \frac{I_D R_S}{V_{GS(off)}} \right)^2$
- $A_v = \frac{g_m R_s}{1 + g_m R_s}$
- $R_{in} = R_G \parallel \left(\frac{V_{GS}}{I_{GSS}} \right)$

COMMON-GATE AMPLIFIER

JFET Self-bias

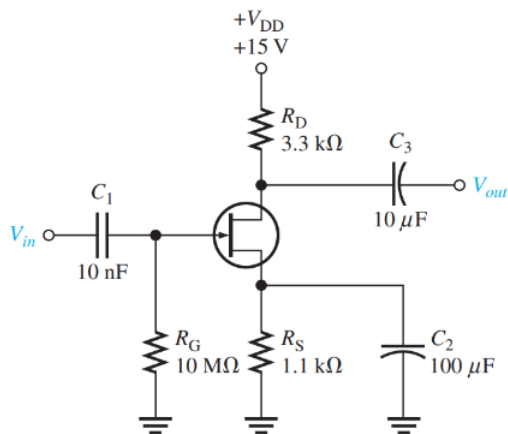


- $I_D = I_{DSS} \left(1 - \frac{I_D R_S}{V_{GS(off)}} \right)^2$
- $A_v = g_m R_d$
- $R_{in} = \left(\frac{1}{g_m} \right) \parallel R_S$

JFET Amplifier Configurations Comparison

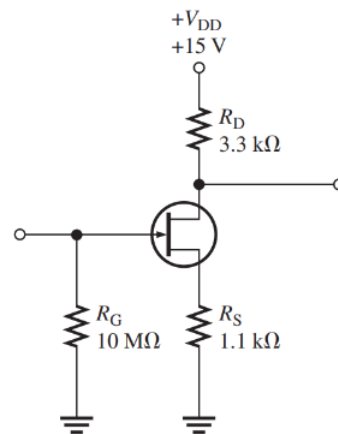
Feature	Common-Source (CS)	Common-Drain (CD / Source Follower)	Common-Gate
Input Terminal	Gate	Gate	Source
Output Terminal	Drain	Source	Drain
Common Terminal	Source	Drain	Gate
Voltage Gain (A_v)	High (Inverting, $\approx -g_m R_D$)	≈ 1 (Non-Inverting, $\approx \frac{g_m R_S}{1+g_m R_S}$)	High (Non-Inv $\approx g_m R_D$)
Current Gain (A_i)	Very High (due to $Z_{in} \rightarrow \infty$)	Very High (due to $Z_{in} \rightarrow \infty$)	≈ 1 (Low current gain)
Power Gain	Highest	Moderate	Moderate
Input Impedance (Z_{in})	Very High (Gate is reverse-biased)	Extremely High (Higher than CS due to feedback)	Low ($\approx 1/g_m$)
Output Impedance (Z_{out})	Moderate to High ($\approx R_D \parallel r_d$)	Very Low ($\approx R_S \parallel 1/g_m$)	Moderate to High ($\approx R_D \parallel r_d$)
Phase Shift	180° (Phase inverted)	0° (In-phase)	0° (In-phase)
Primary Application	General-purpose voltage amplification	Impedance matching / Buffer stage	High-frequency amplification

DC equivalent circuit of common source amplifier



▲ FIGURE 9-5

JFET common-source amplifier.



▲ FIGURE 9-6

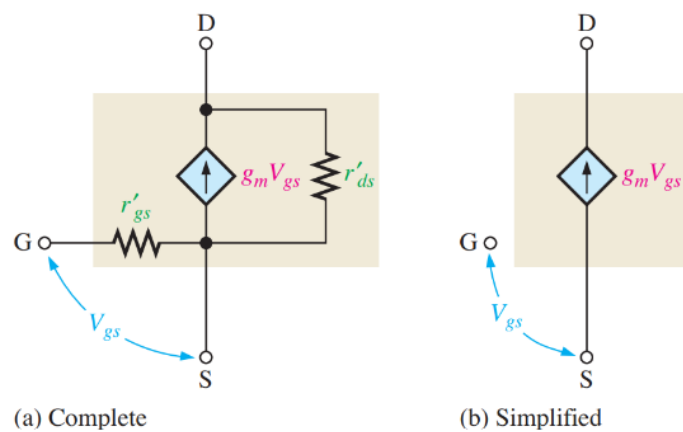
DC equivalent for the amplifier in Figure 9-5.

FET AC Model

An equivalent FET model is shown in Figure 9-1. In part (a), the internal resistance, r'_{gs} , appears between the gate and source, and a current source equal to $g_m V_{gs}$ appears between the drain and source. Also, the internal drain-to-source resistance, r'_{ds} , is included. In part (b), a simplified ideal model is shown. The resistance, r'_{gs} , is assumed to be extremely large so that an open circuit between the gate and source can be assumed. Also, r'_{ds} is assumed large enough to neglect.

► FIGURE 9-1

Internal FET equivalent circuits.



Difference between JFET and MOSFET

2. Key Differences Comparison

Feature	JFET	MOSFET
Full Name	Junction Field-Effect Transistor	Metal-Oxide-Semiconductor Field-Effect Transistor
Gate Insulation	None (uses a reverse-biased p-n junction)	Silicon Dioxide (SiO_2) thin layer
Input Impedance	High (typically $10^9 \, \Omega$)	Extremely High (typically 10^{10} to $10^{15} \, \Omega$)
Operating Modes	Depletion mode only (normally ON)	Depletion mode (normally ON) AND Enhancement mode (normally OFF)
Switching Speed	Moderate	Fast to Very Fast (ideal for high-frequency switching)
Drain Resistance	High	Low (especially power MOSFETs with low $R_{DS(\text{on})}$)
Susceptibility to ESD	Resistant	Highly sensitive to Static Electricity (ESD)
Applications	Low-noise amplifiers, RF mixers, analog switches	Microprocessors, switching power supplies, high-power drivers